

|                                                         |                                                            |
|---------------------------------------------------------|------------------------------------------------------------|
| Program : <b>Diploma in Instrumentation Engineering</b> |                                                            |
| Course Code : <b>5089A</b>                              | Course Title: <b>Industrial Electronics and Drives lab</b> |
| Semester : <b>5</b>                                     | Credits: <b>1.5</b>                                        |
| Course Category: <b>Program Elective</b>                |                                                            |
| Periods per week : <b>3 (L:0 T:0 P:3)</b>               | Periods per semester: <b>45</b>                            |

### Course Objectives:

- To give hands on training on the basic concepts and applications of power electronic devices such as power diodes, SCRs etc.
- To provide experience on building different triggering circuits of SCR and TRIAC
- To experiment with power electronic devices to solve different real life problems.

### Course Prerequisites:

| Topic                             | Course code | Course name                | Semester |
|-----------------------------------|-------------|----------------------------|----------|
| Knowledge of basic Mathematics    |             | Mathematics II             | 2        |
| Basic Knowledge of semiconductors |             | Electronic Instrumentation | 3        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                    | Duration (Hours) | Cognitive Level |
|-----------------|--------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Plot VI characteristics of power electronic devices                            | 15               | Applying        |
| CO2             | Setup different gate triggering circuits of SCR and firing circuits for TRIAC. | 12               | Applying        |
| CO3             | Apply power electronic devices to solve different real life problems.          | 15               | Applying        |
|                 | Lab Test                                                                       | 3                |                 |

**CO – PO Mapping:**

| <b>Course Outcomes</b> | <b>PO1</b> | <b>PO2</b> | <b>PO3</b> | <b>PO4</b> | <b>PO5</b> | <b>PO6</b> | <b>PO7</b> |
|------------------------|------------|------------|------------|------------|------------|------------|------------|
| <b>CO1</b>             | 3          |            |            |            |            | 2          |            |
| <b>CO2</b>             | 3          |            |            |            |            | 2          |            |
| <b>CO3</b>             | 3          |            |            |            |            | 2          | 2          |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| <b>Module Outcomes</b> | <b>Description</b>                                                                    | <b>Duration (Hours)</b> | <b>Cognitive Level</b> |
|------------------------|---------------------------------------------------------------------------------------|-------------------------|------------------------|
| <b>CO1</b>             | <b>Plot VI characteristics of power electronic devices</b>                            |                         |                        |
| M1.01                  | Plot the VI characteristics of SCR                                                    | 3                       | Applying               |
| M1.02                  | Plot the VI characteristics of DIAC                                                   | 3                       | Applying               |
| M1.03                  | Plot the VI characteristics of TRIAC                                                  | 3                       | Applying               |
| M1.04                  | Plot the VI characteristics of UJT                                                    | 3                       | Applying               |
| M1.05                  | Plot the VI characteristics of Power diode                                            | 3                       | Applying               |
|                        | Lab Test – I                                                                          | 1                       |                        |
| <b>CO2</b>             | <b>Setup different gate triggering circuits of SCR and firing circuits for TRIAC.</b> |                         |                        |
| M2.01                  | Setup R-triggering circuit for SCR.                                                   | 3                       | Applying               |
| M2.02                  | Setup RC-triggering circuit for SCR.                                                  | 3                       | Applying               |
| M2.03                  | Setup UJT-triggering circuit for SCR.                                                 | 3                       | Applying               |
| M2.04                  | Setup TRIAC firing circuit using DIAC.                                                | 3                       | Applying               |
| <b>CO3</b>             | <b>Apply power electronic devices to solve different real life problems.</b>          |                         |                        |
| M3.01                  | To control the speed of DC motor using SCR                                            | 3                       | Applying               |
| M3.02                  | To Design and construct an automatic street lamp using optocoupler and LDR.           | 3                       | Applying               |

|       |                                                 |   |          |
|-------|-------------------------------------------------|---|----------|
| M3.03 | To Set up an emergency lamp circuit using SCR   | 3 | Applying |
| M3.04 | To construct a light dimmer circuit using TRIAC | 3 | Applying |
| M3.05 | To construct an inverter circuit using SCR      | 3 | Applying |
|       | Lab Test – II                                   | 2 |          |

#### **Text / Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                                                                           |
|------------|----------------------------------------------------------------------------------------------------|
| R1         | B. L. Theraja, A Text Book of Electrical Eng volume 2, S Chand Publisher.                          |
| R2         | Biswanath Paul, Industrial Electronics and Control, PHI Publisher.                                 |
| R3         | Bimal. K. Bose, Modern Power Electronics and AC Drives, AP Publishers.                             |
| R4         | Joseph Vithayathil, Power Electronics Principles and Applications, McGraw Hill Publisher.          |
| R5         | S. K. Bhattacharya, S. Chatterjee, Industrial Electronics and Control, Technical Education Series. |

#### **Online Resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                                                                                                                                   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://www.tutorialspoint.com/power_electronics/index.htm">https://www.tutorialspoint.com/power_electronics/index.htm</a>                                   |
| 2            | <a href="https://www.electronics-tutorials.ws/power/thyristor.html">https://www.electronics-tutorials.ws/power/thyristor.html</a>                                     |
| 3            | <a href="https://nptel.ac.in/courses/108108077/">https://nptel.ac.in/courses/108108077/</a>                                                                           |
| 4            | <a href="https://www.elprocus.com/electric-drive-types-block-diagram-classification/">https://www.elprocus.com/electric-drive-types-block-diagram-classification/</a> |
| 5            | <a href="https://automationforum.in/t/what-are-electrical-drives/4841">https://automationforum.in/t/what-are-electrical-drives/4841</a>                               |